CAPSTONE PROJECT
LITERATURE SURVEY, GAP ANALYSIS & DATASET FEASIBILITY REPORT

Project Title : Privacy-Preserving Federated Learning System for Missing Person Identification
Domain : Computer Vision, Federated Learning & Biometric Privacy
Reg. No : **1.** 23BCE9603 **2.** 23BCE9585 **3.** 23BCE9746 **4.** 23BCE8540
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Ref	Datasets Used	Methodology / Algorithm	Major Finding	Research Gap
[1]	Manually curated synthetic dataset (780 group images: 240 positive, 540 negative) combining CrowdHuman, WiderPerson, and CelebA In-the-Wild (3–4 reference images per identity).	• **Detection:** Evaluated YOLOv8n to YOLOv12n. • **Embeddings:** ArcFace (InsightFace buffalo_l) and FaceNet (InceptionResNetV1 + MTCNN) with CLAHE preprocessing. • **Matching:** Cosine similarity threshold analysis.	• YOLO–ArcFace attained **97.50% peak accuracy** with near-zero false positives. • YOLO–FaceNet reached **95.42% accuracy** at τ=0.30 but showed higher susceptibility to false alarms. • ArcFace embeddings demonstrated superior inter-class separability under surveillance distortions.	Extend the frame-based architecture to real-time, continuous video tracking utilizing multi-object tracking algorithms like DeepSORT.
[2]	CelebFaces Attributes (CelebA) and Labeled Faces in the Wild (LFW) datasets (500-image subset: 5 train/5 test per identity).	• **Feature Extraction:** InsightFace buffalo_l (512-d). • **Quantization & Protection:** Selected top 96 features quantized via mean thresholding; masked with 32-bit user key via XOR. • **FL & DP:** Federated Averaging (Flower framework) + Gaussian noise Differential Privacy.	• Reached **99.75% accuracy** on CelebA and **99.65%** on LFW. • Attained GAR of **99.53% (CelebA)** / **99.40% (LFW)** with FAR of **0.0002%**. • Validated ISO/IEC 30136 security: resilient against inversion, membership inference, and gradient leakage.	Explore adaptive federated optimization and personalized models for heterogeneous clients; investigate blockchain or SMPC for verifiable aggregation auditability.
[3]	*Not applicable* (Comprehensive review covering face, fingerprint, iris, retina, EEG, ECG, voice, and gait modalities).	• Structured taxonomy evaluating privacy methods across 6 domains: Cryptography (HE, FE, SFE, Lattice-based), Hashing (Bloom filters, LSH), Template Protection (Cancelable, Fuzzy Vaults/Extractors), Computation (SMC, ZKPs, TEE), DP, and Hybrid models.	• Categorized privacy vs. accuracy trade-offs: Cryptography offers highest security but high latency; Hashing is fast but prone to collisions; Template protection ensures revocability; DP prevents re-identification but injects utility-reducing noise.	Optimize techniques for real-time edge/IoT scalability; leverage hardware acceleration (GPUs/FPGAs); integrate machine learning to balance privacy, security, and fairness.
[4]	Pins Face Recognition (17,534 images across 105 celebrities), LFWPeople (1,000 identity classes), MNIST, CIFAR-10, and CIFAR-100.	• **SplitVFed:** Combines Vertical FL (VFL) with Split Learning (SL) across clients, edge servers, and cloud. • **Partial SMPC (P-SMPC):** Additive secret sharing restricted strictly to 1st-layer gradients. • **LDP:** Randomized response applied after feature flattening.	• Achieved **94.68% accuracy** on Pins dataset and **93.61%** with P-SMPC on LFWPeople. • Reduced Membership Inference Attack vulnerability (MIA AUC dropped to **0.507–0.532**, near random guess). • Greatly reduced communication and client device computation overhead.	Test on larger diverse datasets; explore adaptive DP; implement lightweight hardware-assisted cryptographic acceleration; evaluate other modalities and representation inversion risks.

[5]	MEGC 2019 Composite Dataset (442 samples across 68 subjects from SMIC-HS, CASME II, and SAMM).	• **FedMer Framework:** Decentralized Micro-Expression Recognition (MER). • **Preprocessing:** Automatic Clustering (AC, 26 clients) + TV-L1 optical flow features. • **Backbones:** STSTNet, RCN-A, FeatRef. • **Privacy:** RSA-based Homomorphic Encryption + FedAvg + LOSO cross-validation.	• Maintained recognition comparable to centralized models: FedMer_FeatRef achieved **UF1 of 0.7115** and **UAR of 0.7381**. • FedMer_STSTNet reached **UF1 of 0.7093** and **UAR of 0.7316**. • Strong Spearman correlation (~0.98) between federated and non-FL models.	Explore advanced encryption and differential privacy; integrate demographic-aware personalized clustering; evaluate on real-world in-the-wild datasets with occluded micro-expressions.
[6]	Labeled Faces in the Wild (LFW) dataset configured into a Non-IID distribution across clients (5 dominant classes 100 images).	• Evaluated CNN, MobileNetV2, EfficientNet, ResNet50, and a **Transformer-Based Face Recognition (TFR)** hybrid network. • Compared **FedAvg** vs. **FedProx** aggregation. • Applied Differential Privacy (DP) with adaptive clipping and Gaussian noise multipliers.	• TFR paired with **FedProx** achieved the top performance, reaching **89.65% test accuracy** with low average running time. • Higher DP noise led to performance degradation and model divergence in small client setups; sampling 42/2000 clients maintained accuracy.	Optimize federated hyperparameters and network architectures to mitigate client drift and model divergence under strict DP noise budgets in large non-IID deployments.
[7]	CASIA-WebFace (494,414 images from 10,575 subjects for training) and LFW dataset (for face verification testing).	• **BFLFace:** Replaces the central server with a consortium blockchain (FISCO). • **Backbone:** MobileFaceNet trained with ArcFace loss. • **Consensus & Security:** Dynamic committee nodes filter low-quality updates via accuracy thresholds; smart-contract incentive mechanism rewards honest clients.	• Achieved **92.22% verification accuracy**, remaining close to centralized training (94.67%). • Maintained stable ~92% accuracy under 5% and 10% malicious node poisoning attacks (where standard FL failed to ~60%).	Optimize the committee evaluation algorithm and improve execution throughput and incentive fairness across large-scale blockchain networks.
[8]	FER-2013 (35,887 images) and CK+ (~500 images) datasets across 7 emotion classes.	• **Detection & Emotion Classifiers:** Haar Cascade face detector + CNN (14 layers), ResNet-50, and VGG-16. • **FL Training:** Flower framework with FedAvg aggregation. • **Real-Time GUI:** Captures frames per second during ad playback to average emotional engagement.	• Proposed federated CNN achieved **77% training accuracy** and **72.54% test accuracy**, outperforming ResNet-50 and VGG-16. • Successfully preserved user privacy during viewer emotional analysis without transmitting raw photos to central databases.	Implement multimodal federated learning combining facial expressions with speech and natural language processing (NLP) to enrich customer engagement analytics.
[9]	Custom industrial dataset combining factory employee facial recognition attendance logs with Electrostatic Discharge (ESD) wrist-strap/footwear test logs.	• Integrates facial recognition devices and ESD testing machines at the local controller level. • Decentralized model updates trained using the **Pre-FedAvg (Per-FedAvg)** algorithm to handle client-side data imbalance.	• Federated Learning achieved **92% accuracy**, **0.93 F1-score**, and **0.98 ROC AUC**. • Outperformed Differential Privacy by **9.5% in accuracy** (and 8% overall across evaluation matrices) without injecting disruptive noise.	Scale to fully automated industrial environments; leverage aggregated multi-device analytics for factory process optimization (e.g., worker overtime tracking, predictive safety).
[10]	CelebFaces Attributes (CelebA) dataset (1,000 individuals partitioned across 1,000 simulated edge devices).	• VGG-M style CNN classifier with SGD momentum. • Generates negative/imposter samples on-device via (1) CelebA random selection and (2) on-device **GAN** generation. • Secure aggregation evaluated with and without SMC-based Secure Aggregator (SA).	• Federated model without SA achieved an average Equal Error Rate (EER) of **1.66%** (a 26.5% improvement over individual models). • GAN-generated imposters achieved **2.09% EER**. • Adding SA slightly increased EER to **2.72%–2.84%** while guaranteeing parameter privacy.	Optimize secure aggregation protocols and scheduling controls to reduce bandwidth and computational latency associated with on-device GAN training and cryptographic averaging.
[11]	Open-source deep vision benchmark datasets and live edge video sensing streams (NVIDIA Jetson developer testbed).	• Edge-based federated video sensing where raw identifying frames/videos are immediately discarded after processing. • Privacy secured via Additive Gaussian Differential Privacy and Homomorphic Encryption. • Model aggregation via FedAvg and FedOpt.	• Prototyped a minimal edge network testbed preventing video stream leakage. • Verified that communicating AI-framed feature motifs and encrypted model updates prevents leakage without losing recognition saliency.	Evaluate deep networks on commercial IoT hardware; build real-time monitoring/alert dashboards (Power BI/SQL); test across Wi-EDGE urban streetscapes and AR/MR platforms.

[12]	Custom dataset of user facial images captured locally via webcam during job applications.	• **Blockchain:** Ethereum smart contracts (ERC20 tokens) to track application states (Free, Applying, Working) + TScan transaction filter. • **FL:** On-device local face model training (PC vs. Jetson Nano) + Haar Cascade recognition.	• Established tamper-proof, transparent CV records while keeping raw face images on user devices. • Entire recruitment contract execution cost was ~**0.05 ETH**. • PC achieved 13 FPS, 51 ms delay, 65% accuracy (Jetson Nano: 5 FPS, 85 ms delay, 32% accuracy).	Deploy in a real-world recruitment environment; improve facial recognition algorithms to boost FPS and reduce recognition latency on edge hardware.
[13]	CIFAR-10 (AlexNet), Fashion-MNIST (LeNet), BloodMNIST (ResNet-18), and PathMNIST (ResNet-50).	• **CC-APPLF:** Hyperledger Fabric blockchain coordinating dynamic TEE-verified committees. • **Cryptography:** Dual-masking (individual PRG + cancelling pairwise PRG), ElGamal threshold decryption, Shamir Secret Sharing, and Schnorr/BLS threshold signatures. • 4-shard parallel transmission.	• Matched FedAvg accuracy (84.0% vs. 84.5% on CIFAR-10) and outperformed Flamingo by ~1.9%. • Kept communication overhead at O(M), avoiding exponential ECC expansion. • Reduced round training time on PathMNIST (8.9 min vs. 9.8 min for Flamingo).	Expand scalability and dynamic client handling; formulate advanced incentive schemes; integrate with broader Privacy-Enhancing Technologies (PETs) and data lifecycle controls.
[14]	FER-2013 (35,887 in-the-wild facial images) and FERG (55,767 stylized character images across 6 characters).	• **FedAffect:** Collaboratively trains two disjoint models via FedAvg. • **Self-Supervised Learner:** SimCLR-style contrastive learning (ResNet-50V2) on unlabeled private data. • **Few-Shot Classifier:** Relation Network regressing similarity scores (MSE loss) from few labeled samples.	• Achieved **97.3% accuracy** on FERG and **84.9% accuracy** on FER-2013, outperforming centralized baselines. • t-SNE demonstrated that decentralized self-supervised learning learned highly separable, generalizable facial features without data sharing.	Extend the architecture to Non-IID federated distributions; develop automated, end-to-end face localization and extraction from uncontrolled in-the-wild videos.
[15]	VGG-Face dataset (5,000 images depicting 1,000 individuals, 5 images/person).	• **PDID-M:** Hypersphere projection centered on the mean face image. • **SVD-DID:** Three-step SVD modification (Singular Value Zeroing N_Z, Coefficient Averaging low-pass filtering, and Modified Sobel high-pass filtering). • Evaluated against VGG-Face deep CNN; measured image quality using Complex Wavelet SSIM (CW-SSIM).	• Both methods achieved **>90% failure rates** against VGG-Face. • SVD-DID achieved a **92.64% failure rate** while preserving high human visual quality (CW-SSIM c_s = 0.6830), far superior to PDID-M (c_s = 0.2639) and Gaussian blurring (c_s ≤ 0.10).	Develop reversible and less visible face de-identification techniques that preserve privacy against deep neural networks while retaining complete utility for human users.
[16]	Company Employee Dataset: 150 face images from a small-scale company, incrementally augmented to 3,300, 6,300, and 9,450 images. LFW (Labeled Faces in the Wild): 150 randomly selected face images used for non-augmented baseline comparison.	• Model Architecture: Custom multi-layer Convolutional Neural Network (CNN) consisting of 6 convolution layers, 4 max-pooling layers, and fully connected layers. •Data Augmentation: Implemented via Keras ImageDataGenerator using operations such as angular rotation, scrolling, cropping, zooming, and flipping/mirroring, with k-Nearest Neighbors (k-NN) used to fill resultant pixel gaps.	•Accuracy Improvement: Augmenting the dataset substantially increased classification **performance from 50% (original 150 images) to 76%** (9,450 augmented images). •Filter Impact: Angular rotation and brightness filters yielded the greatest positive effect on recognition success; horizontal flipping/mirroring proved more effective than vertical flipping. •Limitation Finding: Excessive angular rotation degraded accuracy due to facial distortion.	Design and test customized filters with optimized parameter ranges to prevent distortion. Evaluate performance across advanced architectures (e.g., AlexNet, ZFNet, GoogLeNet, ResNet, R-CNN) while managing the associated trade-off with increased training duration.

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